Nuclear gluonometry with a tensor-polarized ^{6}Li beam at the Electron-Ion Collider: projected inclusive and coherent cos 2φ measurements

C. Peng and J. Zhou

Physics Division, Argonne National Laboratory

Writing assisted by Claude (Anthropic)

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Abstract. The double-helicity-flip structure function $\Delta(x,Q^2)$ of a spin-1 target is generated at leading twist by gluons alone and has never been measured. We project its measurement at the Electron-Ion Collider with a transversely tensor-polarized ^{6}Li beam, in inclusive deep inelastic scattering, where only the scattered electron is detected, and in coherent diffraction with the ^{6}Li recoiling intact. With the leading-order bag-model moment of Sather and Schmidt ported to ^{6}Li as a labelled scenario, the inclusive cos 2φ amplitudes at the four most sensitive (x, Q^2) bins, x = 0.011–0.14 and Q^2 = 1.1–14 GeV^2, are (0.44–0.95)×10^{-2} at P_{zz} = 0.6, against single-state truth-level statistical errors of (1.4–4.5)×10^{-4} in one EIC year (10 fb^{-1} per nucleon): a 21–44σ measurement per bin whose x dependence discriminates between moment-constrained interpretations, and an extraction of $x\Delta(x,Q^2)$ itself to 2–9% in the best bins. At the reconstructed level, with the spin-state-sorted ratio of m = ±1-rich and m = 0-rich bunches and a hadronic-method y from a PYTHIA 8 final state, the errors are (0.9–3.0)×10^{-4} and the fit closes on the response-folded truth. The coherent channel carries the same modulation with two separable mechanisms — a nuclear-shape term anchored on the polarized-deuteron colour-glass calculation, predicted to flip sign for ^{6}Li, and a flat exotic-glue term — and ^{6}Li's anomalously small quadrupole moment makes it a null test. Its intact recoil keeps the beam rigidity and, with the Yellow Report's beam divergences, cannot be tagged at IP6 with any published optics; the channel defines a far-forward requirement — a dedicated tagging optics or the second interaction region's secondary focus — and for a reference tag with a 0.20 GeV near-beam envelope at the high-acceptance luminosity — which no IP6 optics provides — one year would yield 1.7×10^{7} events and a 5σ sensitivity to an amplitude of 0.9% per unit P_{zz}, against a shape term of 6% and an exotic-glue term of 0.3–1% on the same footing. A tagging optics at IP6 — the horizontal β* raised 46–164× with the pots following the envelope, the vertical plane untouched — recovers a 25–37% tag at 1/6.8–1/12.8 of the luminosity: 2.4–6.2×10^{6} tagged events per year, an 8–11σ measurement of the shape term per year and 5σ on a 1% exotic-glue term in 3–6 years, 4–10× below what a second interaction region's secondary focus would deliver at the same luminosity.

1 · Introduction

A spin-1 target can absorb two units of photon helicity in the forward Compton amplitude. The structure function that measures this, $\Delta(x,Q^2)$, receives no leading-twist contribution from quarks or from bound spin-½ nucleons, so in a nucleus it is direct evidence for glue that is not inherited from the individual nucleons — "exotic glue" in the language of Jaffe and Manohar's nuclear gluonometry [1,2]. It has never been measured for any target; the one prior experimental document is a JLab letter of intent for a ^{14}N target, without a sensitivity projection [3]. The only absolute normalizations available are moment estimates: a bag-model calculation for the spin-3/2 Δ^{++} baryon [4], maximal-transversity scenarios for proton–deuteron Drell–Yan [5], and lattice-QCD matrix elements for the φ meson [6] and the deuteron [7]. The EPIOS white paper on polarized ion beams at the EIC names spin-1 ^{6}Li among the nuclei where a discovery is most likely [8], and the companion physics case [9] argues why ^{6}Li — a polarized deuteron bound to an inert α particle, with A/Z = 2 and deuteron-like spin dynamics in the ring — is the practical choice.

This paper presents the first numerical projection of a Δ measurement for any target. Two channels are considered. In inclusive DIS on a transversely tensor-polarized ^{6}Li beam the scattered electron's azimuth about the alignment axis acquires a cos 2φ modulation proportional to Δ. In coherent diffraction, e ^{6}Li → e′ X ^{6}Li(g.s.), the nucleus stays in its ground state and the modulation of the tagged yield mixes the gluonic term with a nuclear-shape term that the polarized-deuteron calculation of Mäntysaari et al. [10] predicts — and ^{6}Li, with a quadrupole moment fifty times smaller than ^{7}Li's, separates the two. Section 2 states the observable, Section 3 the simulation framework, Section 4 the Δ model, Section 5 the inclusive projections at truth and reconstructed level, Section 6 the coherent channel with the far-forward requirement it imposes, Section 7 the systematics and assumptions.

2 · The observable

For unpolarized electrons on a spin-1 target whose spin axis makes angle θ_m with the beam, the inclusive cross section per spin projection m is [1,2]

$$\frac{d\sigma^{(m)}}{dx\,dy\,d\phi}=\frac{2y\alpha^2}{Q^2}\left[F_1+\tfrac{2}{3}a_m b_1+\frac{1-y}{xy^2}\left(F_2+\tfrac{2}{3}a_m b_2\right)-\frac{1-y}{y^2}\,c_m\sin^2\theta_m\,\Delta(x,Q^2)\cos 2\phi\right]$$

$$a_m=\tfrac{1}{4}\,c_m(3\cos^2\theta_m-1),\qquad c_m=3|\lambda_m|-2\;\rightarrow\;(1,\;-2,\;1)$$

with φ the azimuth of the lepton plane about the virtual photon, measured from the transverse projection of the spin axis. Averaged over a fill, c_m becomes the tensor polarization $P_{zz} = p_+ + p_- - 2p_0$. We write the b_1, b_2 terms with the sign of [2] as transcribed in the companion physics case [9]; in the Hoodbhoy–Jaffe–Manohar convention as applied by HERMES and by Cosyn et al. [11] they carry the opposite sign, which affects only the sign of A_{zz} relative to b_1 and the constant term of the spin-state ratio of §3.3, nothing in the Δ sector. At leading order Δ is generated by the gluon transversity distribution δG through the $O(\alpha_s)$ photon–gluon convolution

$$\Delta(x,Q^2)=-\frac{\alpha_s(Q^2)}{2\pi}\left(\sum_q e_q^2\right)x^2\int_x^1\frac{dy}{y^3}\,\delta G(y,Q^2)$$

in the sign convention of Ref. [6]. For a transverse spin axis (θ_m = 90°) the measured amplitude and its inversion to the structure function are

$$A^{\cos 2\phi}=-\frac{1-y}{y^2}\,\frac{\Delta}{D_\phi},\qquad D_\phi=F_1+\frac{1-y}{xy^2}\,F_2,\qquad \hat{\Delta}=-\hat{A}\,\frac{y^2 D_\phi}{1-y}$$

Three properties of this relation shape the analysis. First, with $F_2 = 2x(1 + R)F_1$ the inversion becomes $\Delta/F_1 = -\hat{A}[2(1 + R) + y^2/(1 - y)]$, so at the $y \lesssim 0.03$ of the sensitive region F_2 cancels and $R = \sigma_L/\sigma_T$ is the only structure-function input that reaches Δ/F_1. Second, the amplitude is exactly linear in Δ (§5.3). Third, the target-mass corrections are small: at the four bins of §3.4, $\gamma^2 = 4M^2x^2/Q^2 \le 0.0057$ with the free-nucleon M = 0.9383 GeV that a per-nucleon x requires (the bound-nucleon mass would lower it by 1.0%), so the approximation set of the simulation — $g_2 = g_2^{WW}$, and until 2026-08-29 $A_\parallel = D\cdot g_1/F_1$ with the γ^2-suppressed term dropped — is adequate. The exact longitudinal kernel $A_\parallel = D_\gamma(A_1 + \eta A_2)$ is now the default (`InclusiveKernel(target_mass=True)`; `target_mass=False` restores the massless limit); at the $y \lesssim 0.03$ of these bins it collapses to a multiplicative $(1 + \gamma^2)$ on $A_\parallel$, independent of g_2 — ≤ 0.6% here, ≤ 0.5% at the top configuration and ≤ 2.5% at the low one, whose Q^2 = 1.14 GeV² spot sits at x = 0.089 — and the $W^2 \ge 10$ GeV² cut bounds it by $\gamma^2 \le M^2/(W^2_{min} - M^2) = 0.097$ anywhere in the acceptance. No Δ number reported here is built from $A_\parallel$. One finite-γ effect is physical rather than a labelling matter: the tensor-polarized b_1–b_3 terms feed the cos 2φ amplitude at $O(\gamma^2)$, bounded by $1.15\,\gamma^2\,b_1/F_1$ (the full Cosyn et al. Eq. 17e combination, ≈ 6.9× the $\gamma^2 b_1/6$ of the b_1 term alone) [11]; for a HERMES-like b_1 this is ≤ 1% of the projected amplitudes at the four bins and is computable from the same data's A_{zz}.

The size of Δ is unknown to within an order of magnitude. The bag-model moment of Sather and Schmidt, $\int x\,\Delta\,dx = -0.012\,\alpha_s$ with its Q^2 evolution but no x dependence [4], puts the asymmetry at the per-mille level; the explicit $\alpha_s(Q^2)$ is the leading-order Wilson coefficient of the photon–gluon box, so the $O(\alpha_s)$ suppression relative to F_1 is model-independent while −0.012 is the bag estimate. Lattice QCD adds two results, both at unphysical pion masses: the bare φ-meson matrix element A_2 = 0.23(2)(5) at m_π = 450 MeV [6], which converts to $\int x\,\Delta|_\varphi \approx -0.006\,\alpha_s$ (our conversion, unrenormalized), and the NPLQCD deuteron δG moment of −0.010(3), bare at $m_\pi \approx$ 806 MeV [7] — a transversity-to-momentum ratio roughly ten times below the φ's, implying the opposite sign for the deuteron's Δ moment. Percent-level asymmetries arise only in the maximal-transversity ($\Delta_T g = \Delta g$) Drell–Yan scenarios their authors call optimistic [5]. Nothing exists for ^{6}Li or any A > 2 nucleus beyond a ΔΔ-isobar estimate for the deuteron [12], so the literature brackets the nuclear Δ in both directions:

suppressed, as in the NPLQCD deuteron, or enhanced by binding in analogy with the EMC effect [8]. The projections therefore carry the moment-constrained model of §4 together with the scenario band $\Delta/F_1 \in 10^{-3}$–10^{-2}.

3 · Simulation framework

3.1 · Beams, polarization and luminosity

EIC ions are γ-matched to the proton rather than rigidity-scaled, because the two rings must share a revolution period, so the ^{6}Li menu is 40.8 GeV/u and 99.5–137.5 GeV/u, the top value being the rigidity cap 275 GeV × Z/A [8,13]. The projections are made at the mid configuration, e(10 GeV) × ^{6}Li(99.5 GeV/u), $\sqrt{s_{NN}}$ = 63 GeV, with the low (5 × 40.8) and top (18 × 137.5) configurations used where the hadronic reconstruction or the far-forward acceptance depends on the energy. The in-ring tensor polarization is a placeholder P_{zz} = 0.6 (the ANL source targets $P_{zz} \geq 0.8$ [9]; transport to the ring is an open accelerator question, and every statistical error scales as $1/P_{zz}$), the alignment axis is the vertical — the stable spin direction of the arcs, native at the interaction point — and the electrons are unpolarized. Luminosity is quoted per nucleon: 10 fb^{-1}/u, "one EIC year" in the Yellow Report's accounting (30 weeks at 10^{33} cm^{-2} s^{-1} with 60% efficiency [13]), and 100 fb^{-1}/u for the ten-year programme, all of it in the transverse-tensor fill. No lithium luminosity has been published; the yields scale linearly with whatever the machine delivers. The inclusive and coherent measurements want different far-forward optics — the Yellow Report divergence for the first, the tagging optics of §6.1 at 1/6.8 to 1/12.8 of the luminosity for the second — and each projection assumes the full luminosity of the year in its own, so the two reaches are alternatives rather than a sum; the share the run plan actually gives each is a programme decision, priced in plans/07 WP2, and every statistical error below scales as one over the square root of it, while a luminosity quoted as a reach in fb^{-1}/u stands where it is and the years needed to accumulate it grow as one over the share. What does not follow that law is anything in an error bar that is not statistics: the full one-year Δ bars of Report 2 [16] carry the spread over unfolding priors, which no luminosity shrinks, and it has to be separated out before a share is applied.

3.2 · Generator and structure functions

No public generator produces polarized e+A events for A > 1: BeAGLE is unpolarized, DJANGOH and PEPSI are nucleon-level and longitudinal-only. The generator used here, polligen, implements the missing component — the spin-density matrix ⊗ structure-function master formula with spin-correlated sampling — and takes the hadronic final state from a PYTHIA 8 production of 8 M events over the three beam configurations. It is validated against the analytic asymmetry formulas and by pseudo-experiments, including unbiased cos 2φ recovery under φ-acceptance holes; a full-luminosity pseudo-experiment draws each φ-bin count from a Poisson distribution with the exact expected yield, which is statistically identical to event-level sampling.

Structure functions are toy parameterizations with PDF-grid backends (CT18NLO, EPPS21, NNPDFpol11) behind one interface, so rates hold to a factor ≈ 1.5 (the toy F_2 is within ±37% of CT18 over the acceptance) and asymmetry shapes are scenarios by construction. $R = \sigma_L/\sigma_T$ is a placeholder of the right magnitude, 0.14–0.18 at the four bins of §3.4, in every projection below; the published SLAC/E143 R1998 world fit [14], all three of its functional forms, sits behind the same hook, and switching to it moves Δ/F_1 by +18, +19, +8 and −4% at the four bins, with a further 0.5–6% from the spread of the fit forms — a correction to apply rather than a band, and the systematic on every Δ quoted below until it is.

3.3 · Estimators

The figures use a single spin state: a binned least-squares fit of $N(\varphi') = C(1 + A \cos 2\varphi')$ rather than the $2\langle\cos 2\varphi'\rangle$ moment, which realistic φ holes bias by more than the signal while the fit stays unbiased, with $\delta A = \sqrt{2/N}/P_{zz}$ validated by pseudo-experiments and no cos φ′ term, since the m-symmetric tensor fills set it to zero by construction. The azimuth φ′ is the covariant one built from the four-vectors [15], equal to $\varphi_e - \varphi_S$ for a massless target and within 5 mrad of it for ^{6}Li, and the 25 mrad crossing angle changes only odd harmonics of the electron azimuth because the ePIC axis is the electron beam.

The measurement itself cannot use a single state: the fit then also measures the detector's own cos 2φ′ acceptance harmonic divided by P_{zz}, and a 3% harmonic aligned with the vertical spin axis fakes an amplitude of 0.05, five to eleven times the signal. The estimator is therefore the spin-state-sorted ratio of m = ±1-rich (P_{zz} = +0.6) and m = 0-rich (−1.2) bunches. It cancels any acceptance common to both states bin by bin, turns relative-luminosity errors into a constant, and gives, with N the total of both states, $\delta\hat{A} = 2\sqrt{(2/N)}/(P_+ - P_0) = 0.67\times$ the single-state error if the m = 0-rich fill reaches $|P_{zz}| = 1.2$. The cancellation of the acceptance, however, holds only for an acceptance common to both states: for two fills the bias is $(\varepsilon_2^+ - \varepsilon_2^0)/(P_+ - P_0)$ at any luminosity split, so a 10^{-3} difference of the cos 2φ′ harmonic fakes 5.6×10^{-4}, 6–13% of the projected amplitudes and 2–6 one-year statistical errors. Bunch-by-bunch alternation, or 10^{-4} stability of the harmonic between fills, is thus a requirement of the measurement rather than a refinement; with it, a relative-luminosity error between the states enters only as a constant.

3.4 · Acceptance and binning

Rates use per-nucleon luminosity, F_2^A/A, and scattered-electron cuts $|\eta| \le 3.5$, $E' \ge 0.5$ GeV, $0.01 \le y \le 0.95$ and $W^2 \ge 10$ GeV², which accept 5.2×10^9 events in one year (Figure 1). The four displayed (x, Q²) super-bins are selected by statistical significance within Q² bands — an unconstrained ranking concentrates at the lowest Q², whereas the Q² lever arm provides the consistency check against power-suppressed contributions — and each merges 3×3 cells of the 40×30 analysis grid (3×2 for the two Q² = 1.14 GeV² bins, clipped at the grid's Q² = 1 GeV² edge). They sit at (x, Q²) = (0.028, 1.14), (0.011, 1.14), (0.071, 3.13) and (0.141, 14.3 GeV²), i.e. at y = 0.010, 0.026, 0.011 and 0.026: the amplitude grows as $(1 - y)/y^2$, so the sensitivity lives at the lowest y the detector can reconstruct, which is what makes the kinematic reconstruction of §5.2 the central detector question.

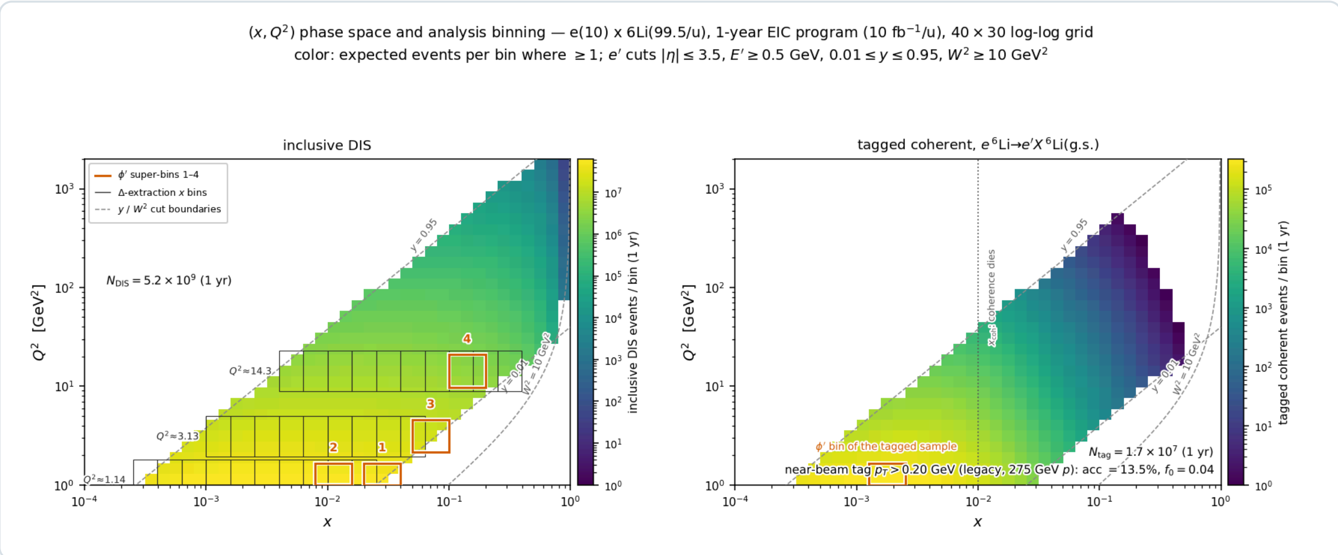

Figure 1 — phase-space coverage and analysis binning at e(10 GeV) × ⁶Li(99.5 GeV/u), one EIC year, 40×30 log-log grid; colour: expected events per bin where ≥ 1; dashed: y and W² cut boundaries. **Left, inclusive DIS**: boxes 1–4 are the φ′ super-bins of Figure 2; grey combs are the three Q² slices (Q² × [1/1.6, 1.6]) whose merged x-bin pairs give the data points of the amplitude-versus-x panel and of the Δ extraction (Figures 2 and 3), retained where $N \ge 10^3$ and $\delta A \le 8\times10^{-3}$. **Right, tagged coherent channel**: the same map times $f_{coh}(x)$ and the tag acceptance of §6.2, almost entirely at x < 0.03; the box is the super-bin of Figure 5(d).

4 · The Δ model

The default model, A, is an ansatz normalized to the bag moment applied to the per-nucleon Δ of ⁶Li:

$$\Delta_A(x,Q^2) = \tfrac{1}{3}\,A\,\alpha_s(Q^2)\,F_1(x,Q^2)\,\frac{x^a(1-x)^b}{N_{peak}}, \qquad \int_0^1 x\,\Delta(x,\langle Q^2\rangle)\,dx = -0.012\,\alpha_s \Rightarrow A = -0.292$$

with (a, b) = (0.7, 3). The factor 1/3 restricts the exotic glue to the polarized deuteron-like pair while the whole-nucleus P_{zz} remains a separate factor, so nothing is double counted; the cluster picture would give ≈ 0.81 instead, an upside of ×2.4 that is not taken. The constraint is imposed at the rate-weighted ⟨Q²⟩ = 4.47 GeV² of the accepted phase space, and because F_1 evolves the realized moment varies by a few per cent across the plotted Q² slices; α_s comes from the CT18NLO table. Model B imposes the same constraint without the F_1 factor and yields amplitudes a factor 3–56 lower at the four bins (20–56 in the Q² = 1.14 GeV² bins, 3–8 at the higher Q²); the A-versus-B spread is the dominant ansatz systematic, and both curves appear on every plot. Porting a single-hadron bag moment to a per-nucleon nuclear structure function has no precedent, so normalization and x dependence are both scenario choices, bracketed by model B from below and possible nuclear enhancement from above.

5 · Inclusive projections

5.1 · Truth level

Figure 2 shows the φ′ pseudo-data in the four super-bins with one-year statistics and the extracted amplitude versus x along the $Q^2 = 1.14$ GeV² slice. The injected amplitudes are 7.4, 4.4, 9.5 and 9.5 $\times 10^{-3}$ against one-year errors of 1.7, 1.4, 2.8 and 4.5 $\times 10^{-4}$ — 44, 31, 34 and 21σ — and ten-year errors of 0.55, 0.45, 0.87 and 1.4 $\times 10^{-4}$; the fitted values agree with the injected ones within 1.5 standard errors. Under the moment constraint the measurement is not a null test: the model-A-versus-B spread is resolved by the x dependence of the amplitude, and a flat amplitude of 3×10^{-3} per unit P_{zz} would remain a 3–11σ effect per bin even at $P_{zz} = 0.3$. At ten-year precision the statistical errors sit below the polarimetry scale systematic (3% of an amplitude of 10^{-2} is 3×10^{-4}), so the ten-year measurement is systematics-limited.

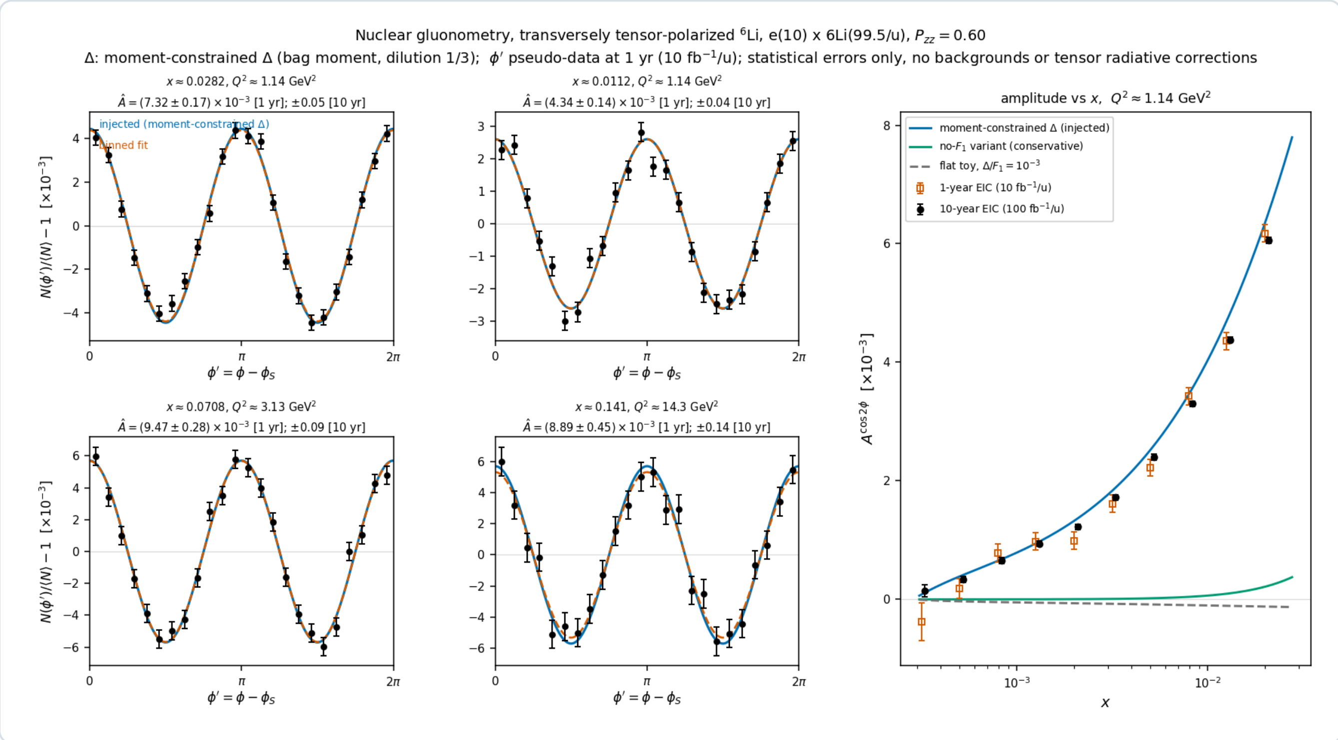

Figure 2 — inclusive gluonometry at e(10 GeV) × ⁶Li(99.5 GeV/u), P_{zz} = 0.60, model A with dilution 1/3. Left 2×2: φ′ pseudo-data with one-year statistical errors in the four super-bins (Q^2 = 1.1 → 14 GeV², x = 0.011–0.14); injected curve (blue), binned fit (vermillion); annotations give δA for both programmes. Right: extracted amplitude versus x with one-year (open) and ten-year (filled) errors against the model A, model B and flat-10^{-3} curves.

Figure 3 unfolds the same pseudo-measurements to the structure function, $x\Delta(x,Q^2)$ per nucleon at the three Q^2 slices of Figure 1, with independent one- and ten-year draws and the model bin-centering factor $K = \Delta_{\text{model}}(x_c, Q^2_c)/A_{\text{model, bin}}$; the area under each curve is the ported bag moment (÷3 dilution). The best bins reach Δ = −0.135 ± 0.003 at (x, Q^2) = (0.02, 1.14), −0.070 ± 0.002 at (0.05, 3.13) and −0.0047 ± 0.0004 at (0.32, 14.3) in one year, i.e. 2–9% relative, and three times better in ten.

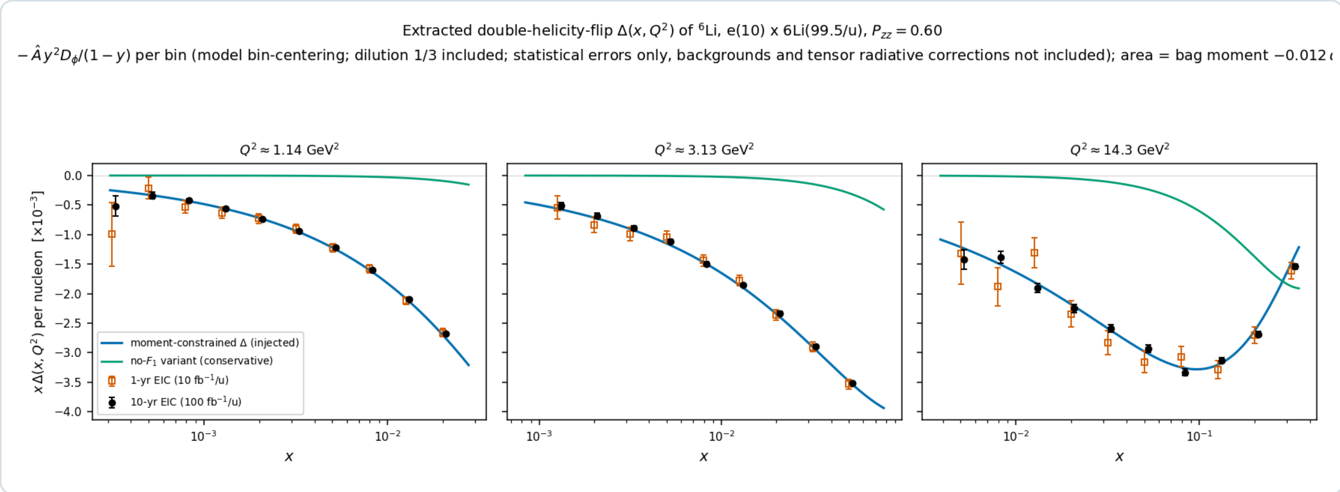

Figure 3 — the extracted $\Delta(x,Q^2)$: the amplitudes of Figure 2 unfolded to $x\Delta$ per nucleon at the three highest-significance Q^2 slices (model bin-centering, $\delta\Delta = \delta A\cdot|\partial\Delta/\partial A|$), with independent one-year (open) and ten-year (filled) pseudo-measurements against the model A and model B curves.

5.2 · Reconstructed level

Two of the four super-bins sit at $y \approx 0.01$, where the electron method alone gives $\delta y/y \approx 1.1$–1.2 and the bin is not reconstructible, and the other two at $y \approx 0.026$, where it gives 0.46; x must come from a hadronic-method y. The reconstructed-level pseudo-experiments of the companion note [16] pass every event through the lab frame, EMCal energy and track-angle smearing, the head-on transformation, the electron-method Q^2, a hadronic y, the mixed-method x, the analysis cuts on reconstructed quantities, an η-dependent electron identification efficiency and the covariant φ′, then form the spin-state ratio of the two fills with a 3% cos 2φ′ and 2% cos φ′ acceptance harmonic and a 10^{-3} relative-luminosity offset switched on. Table 1 gives the result with the documented 25% Gaussian stand-in for the hadronic y — the ePIC kinematic-fit study's smearing and the ATHENA value at $y \approx 0.01$ [17] — and with the y of the PYTHIA 8 final state through the hadron-side detector response (Σ method, 50 MeV of calorimeter noise), the hadronic scale calibrated from the same simulation as an analysis would, in bins of the reconstructed (x, Q^2) [16].

SUPER-BIN	25% HADRONIC-Y STAND-IN			PYTHIA 8 FINAL STATE, Σ METHOD			ΔÂ (1 YR)	ΔÂ (10 YR)
(X, Q²)	PURITY	EFFICIENCY	D	ΔY/Y	PURITY	D	[10^{-4}]	[10^{-5}]
0.028, 1.14	0.66	0.42	0.92	0.32	0.56	1.00	1.2	3.7
0.011, 1.14	0.63	0.59	0.99	0.22	0.59	1.06	0.9	2.9
0.071, 3.13	0.69	0.38	0.90	0.29	0.64	0.95	1.6	5.2
0.141, 14.3	0.69	0.66	0.96	0.15	0.75	0.98	2.9	9.3

Table 1 — the four super-bins at the reconstructed level (mid configuration, spin-state ratio of the +0.6 / −1.2 fills). Purity: fraction of the reconstructed bin's rate that originates in the true bin; efficiency: fraction of the true-bin rate reconstructed in the bin after electron identification; D: rate-weighted, φ′-diluted amplitude of the events in the reconstructed bin over that of the true bin; δy/y: 68% half-width of the Σ-method resolution at the spot. The PYTHIA columns have the hadronic scale calibrated in reconstructed (x, Q^2) bins; uncalibrated they read purity 0.42 / 0.53 / 0.49 / 0.68 and D = 0.79 / 0.84 / 0.83 / 0.95. The statistical errors are the same for both hadronic-y models to within 7%.

Three conclusions follow. The fit closes on the response-folded truth within the precision of the test: the eight pulls against the reconstructed-bin truth lie within 2.3 standard errors while the acceptance harmonic and the luminosity offset are on, and the sin 2φ′ coefficients are consistent with zero. The errors are 0.59–0.69 of the single-state truth-level values, because the 1.5× gain of the m = 0-rich fill outweighs the reconstruction efficiency of 0.38–0.66 — presuming the source delivers m = 0-rich bunches at $|P_{zz}| = 1.2$ with the same purity as the ±1 fill at 0.6. And the real final state costs purity rather than statistics: only 80–92% of the hadronic $E - p_z$ sum is within the acceptance and above threshold at the four bins, 70–85% through the full response — the rest escapes forward beyond the calorimeters (7–17%, the target-fragmentation side of a W ≈ 6–10 GeV system) or lies below threshold — a 13–27% scale bias on y_Σ that an analysis calibrates from its simulation at the reconstructed point. Calibrated, the purities are 0.56–0.75 and the reconstructed-bin amplitude within 7% of the true-bin value (Table 1); uncalibrated they would be 0.42–0.68 and 5–21% low, and a 1% residual scale error moves Â by 0.1–0.6% [16]. The noise floor decides the resolution below $y \approx 0.02$ — at the mid configuration the Σ-method $\delta y/y$ at $y = 0.005$ runs 0.22 → 0.33 → 0.54 → 1.01 for 0 → 25 → 50 → 100 MeV of calorimeter noise on a hadronic $E - p_z$ sum of 0.2–0.5 GeV, a regime no published ePIC study covers. Each configuration selects its own four bins: at 5 × 40.8 and 18 × 137.5 they resolve to 0.39 / 0.23 / 0.24 / 0.11 and 0.23 / 0.19 / 0.21 / 0.18, so the low-x bins improve with beam energy (the hadronic $E - p_z = 2E_e y$ grows with it) while the (0.141, 14.3 GeV²) bin degrades, 0.11 → 0.15 → 0.18, and is best measured at the low configuration. Unfolded to Δ, the best reconstructed bins reach statistical errors $\delta\Delta = 2.4\ (2.3) \times 10^{-3}$ at $Q^2 = 1.14$ GeV² and $1.2\ (1.2) \times 10^{-3}$ at 3.13 GeV² in one year with the stand-in (PYTHIA) y, at purities 0.56–0.57 (0.44–0.50).

5.3 · Bin centering

Correcting the reconstructed amplitude to a point value of Δ with the model's own K is circular. Because the amplitude is exactly linear in Δ, the response can instead be applied as a linear operator and a Δ(x) shape fitted through it per Q² slice on a bounded grid. Correcting the same pseudo-data with either of two deliberately wrong priors, the model dependence of K at the four super-bins falls from (−4.1, +8.0, −5.5, +4.9)% to (−1.5, −1.8, −0.3, +0.5)% with the model B prior and from (+8.6, +2.5, +7.3, +3.2)% to (+4.6, +3.0, +1.8, −0.9)% with the flat toy, and over the plotted bins of Figure 3 from up to 29% to ≤ 6%. The Δ error bar then carries four terms — statistics, the shape fit, the response Monte Carlo (0.15–1.5% per bin) and the spread over priors the shape family cannot absorb (1–8%) — of which the prior spread dominates in the best-measured bins, where the one-year statistics are ≲ 4%, and statistics elsewhere.

6 · The coherent channel

6.1 · Detecting the intact recoil

In coherent diffraction the recoil retains the beam's magnetic rigidity — A/Z = 2 exactly, and the diffractive momentum loss $x_P < 0.01$ changes it by less than 1% — and emerges at $\theta = p_T/p \lesssim 1$ mrad, inside the beam. Of the far-forward system at IP6 — the B0 spectrometer, the off-momentum detectors, the Roman Pots and the zero-degree calorimeter, with the acceptance windows of Refs. [18–20] — only the pots can see such a recoil, and only where it clears both the 10σ beam envelope and the pots' own aperture. Both are constraints on the *angle* at the interaction point, so with an exponential t distribution of slope B (≈ 50 GeV⁻², §6.2) the tagged fraction is

$$\varepsilon_{\rm tag} = \exp\left(-B\, p_{T,\rm cut}^2\right), \qquad p_{T,\rm cut} = \theta_{\rm cut} A\, p_u, \qquad \theta_{\rm cut} = \max(10\,\sigma_\theta,\ \theta_{\rm aperture})$$

an exponential in the square of the beam divergence times the beam momentum. Table 2 evaluates it with B = 50 GeV⁻² at the three ⁶Li configurations, using the Yellow Report's divergences per configuration and optics — a γ-matched ion has the proton's βγ and hence the proton's divergence at equal normalized emittance, and pays a factor √2 only where it is rigidity-capped, i.e. at the top configuration [13] — and the pots' aperture for an intact ⁶Li, measured by tracking one through the ePIC far-forward geometry: the boundary is $|\theta_x| \gtrsim 2.50 / 1.51 / 0.53$ mrad in the 5 × 41 / 10 × 100 / 18 × 275 optics against $|\theta_y| \gtrsim 0.92$–2.12 mrad, and no vertical acceptance at all at 5 × 41, because the transport images an interaction-point angle onto the pot plane with a horizontal lever R_{12} = 19.24 / 21.25 / 29.97 m against a vertical R_{34} = 3.35 and 2.93 m, a factor 6.3 at 10 × 100 and 10.2 at 18 × 275 [16]; the scan was made at the proton rigidity of each optics and is applied in angle to the γ-matched beam, which assumes the far-forward magnets scale with the stored rigidity.

⁶Li configuration	$p_{\rm ion}$	σ_Θ H/V, high acceptance [mrad]	$\varepsilon_{\rm tag}$, 10σ envelope	Pot aperture $\lvert\theta_x\rvert$	$\varepsilon_{\rm tag}$, aperture	Tagging optics: σ_Θ, β* factor
5 × 40.8	245 GeV	220 / 380	5×10⁻⁷	2.50 mrad	9×10⁻¹⁰	58 µrad, ×14.5
10 × 99.5	597 GeV	180 / 180	8×10⁻²⁶	1.51 mrad	2×10⁻¹⁹	24 µrad, ×58
18 × 137.5	825 GeV	92 / 92	4×10⁻¹³	0.53 mrad	1×10⁻⁵	17 µrad, ×29

Table 2 — the coherent tag at IP6 with published inputs. Divergences: Yellow Report Table 10.1 with the γ-matched species step [13]; at 41 GeV the high-acceptance and high-divergence optics are identical, the only configuration where the Yellow Report offers no separate high-acceptance option. Aperture: the rectangular $|\theta_x|$ / $|\theta_y|$ cutout integrated over the two-dimensional Gaussian, not the circular $\exp(-B\,p_T^2)$ of the displayed formula; the aperture column is the re-measurement of 2026-08-28 in the current ePIC geometry (`epic-main` at git 9aaa2969, 16 mm modules and per-energy insertions), one event per scan point, no beam envelope, on 0.05 mrad angular ladders at four azimuths; the 2.0 / 1.35 / 1.03 mrad it carried before were measured in a September-2024 layout. The 5 × 41 entry stands on the ePIC baseline proton lattice for that ring setting; the Z/A = 0.5 file at the ^{6}Li fill point gives 1.61 mrad instead — 48 mm / 29.81 m, the threshold the code applies, against a first hit at 1.60 mrad on +x and 1.70 on −x — a ×0.64 systematic on that row alone. Tagging optics: the divergence that puts the 10σ envelope at $p_T = 1/\sqrt{B} = 0.14$ GeV, where $L \times \varepsilon_{tag}$ is maximal and $\varepsilon_{tag} = 1/e$, and the β* increase over the high-acceptance optics it implies; Figure 4 prices the planar-pot version.

The conclusion is unambiguous: with any published optics the coherent intact-^{6}Li channel does not exist at IP6, at any energy — the largest entry in the table is 10^{-5} — although which of the two constraints delivers the verdict changed hands with the re-measurement: at 5 × 40.8 the silicon now binds, at 2.50 mrad against a 2.20 mrad envelope, and at 18 × 137.5 the beam does, the aperture having halved to 0.53 mrad against an envelope of 0.92. It exists under two conditions. A *tagging optics*, with β* raised until the envelope sits at $p_T = 1/\sqrt{B}$, where luminosity times acceptance is maximal, gives $\varepsilon_{tag} = 1/e$ at every configuration; the 1/e column of Table 2 is the circular-isotropic derivation, and priced for planar pots below the optimum is a horizontal-only de-squeeze, within the range of the forward-physics optics demonstrated at the LHC — its size, and the approach it then demands of the pots against a silicon edge at 2.50 / 1.51 / 0.53 mrad, are derived below; the two levers are multiplicative and the second is the near-beam question of Report 4 [21]. Alternatively, the *secondary focus* of the second interaction region lets pots at z = 44.0/45.5 m reach $p_T \approx 0$, with published intact-recoil efficiencies of 47% (d), 32% (^{3}He), 29% (^{4}He) and 17.8% (^{7}Li) at top energy, rising steeply at lower energies [22]; no ^{6}Li case is published and our interpolation is ≈ 20%.

Pricing the tagging optics. The β* lever is a machine lattice setting, not a far-forward one — it is exactly what separates the Yellow Report's high-acceptance optics from its high-divergence optics — and it is paid for in luminosity: $\sigma_\theta \propto 1/\sqrt{\beta^*}$ while $L \propto 1/\beta^*$ per plane, which the Yellow Report's own optics pairs reproduce to 5% (10.0 against 3.14×10^{33} cm^{-2}s^{-1} at 275 GeV for a divergence ratio 119/65) [13]. Because a recoil escapes through the horizontal gap between the pots or through the vertical one, only the plane it escapes through needs the tagging β*: Figure 4 therefore de-squeezes the horizontal plane alone, holds the vertical at its high-acceptance value and pays $L/L_{HA} = 1/\sqrt{r}$ rather than 1/r, with the envelope treated as the planar pots see it — a rectangle $10\sigma_h \times 10\sigma_v$ in angle, the dispersion at the pots (D ≈ 0.3 m with the Yellow Report Δp/p) added to the horizontal side — for pots either following the envelope or fixed at the aperture measured today. With the pots fixed, the acceptance peaks at 1.2×10^{-5} over β*: the far-forward side must move with the optics. With the pots following, the yield peaks at $r = \beta^*_x/\beta^*_{x,HA}$ = 46, 164 and 89 at the three configurations, where ε = 0.37, 0.25 and 0.33, the horizontal envelope sits at 0.36, 0.19 and 0.12 mrad (0.09–0.11 GeV) and the luminosity at 1/6.8, 1/12.8 and 1/9.5 of the high-acceptance value; de-squeezing both planes, the naive reading of Table 2, would cost 1/25–1/71 for a fifth to a quarter of the yield. Taking the placeholder of §3.1, 10 fb^{-1}/u per year, as the high-acceptance luminosity, the tagging optics yields 2.4×10^6, 2.4×10^6 and 6.2×10^6 tagged events per year — 2.5×, 6.9× and 3.9× below the 0.20 GeV reference tag at that luminosity — and the best super-bin of §6.2 reaches 5σ floors of 1.7%, 2.3% and 1.6% per unit P_{zz} per year. In the optics' own window, ⟨|t|⟩ = 0.033–0.039 GeV², the shape term of §6.3 is 3.3–3.9% per unit P_{zz} — a 9σ, 8σ and 11σ measurement per year — and a 1% exotic-glue term reaches 5σ in 3.0, 5.5 and 2.6 years. The top configuration is the most productive at equal luminosity, with four times the coherent events of 5 × 41. If the second interaction region delivered the same luminosity as the high-acceptance optics, its secondary focus at our interpolated ≈ 20% would still be worth 4×, 10× and 6× the IP6 optimum in yield (Figure 4, dash-dotted line); the factor scales with L_{IR8}/L_{HA}, which is not published, and the published efficiencies rise toward lower energies [22]. Two assumptions carry these numbers and are stated as such: the electron β* must be raised in step to keep the spot sizes matched, and the far-forward transport of the de-squeezed lattice must remain parallel-to-point at the pots ($R_{11}\sigma^* \ll R_{12}\sigma_\theta$, as in TOTEM's high-β* optics) for the envelope to be $10R_{12}\sigma_\theta$ at all — true of the present lattice at the Yellow Report optics, and the first thing to establish for a de-squeezed one.

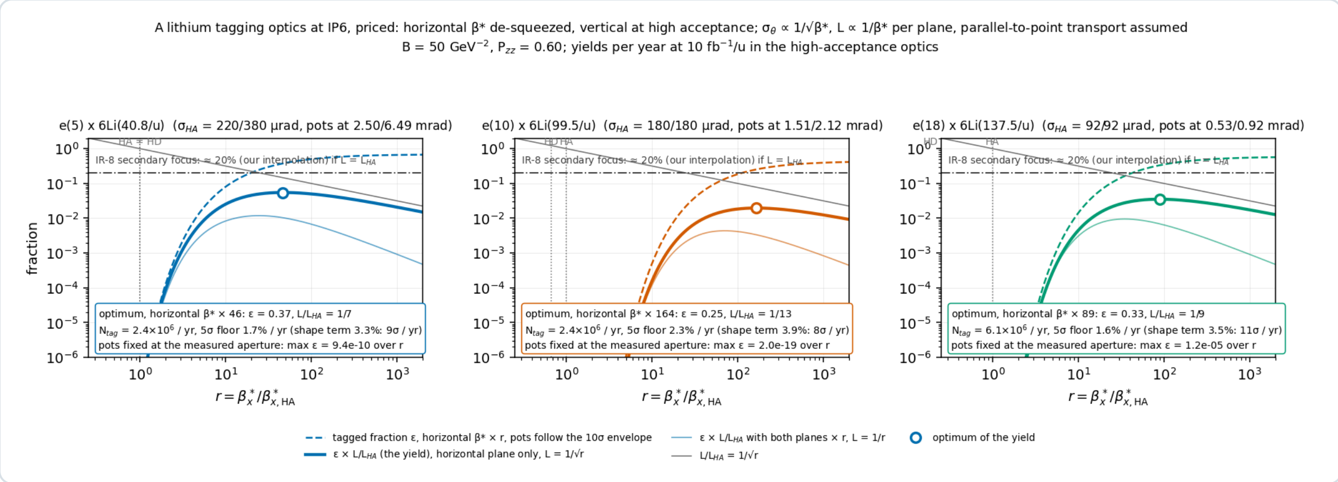

Figure 4 — a lithium tagging optics at IP6, priced. Per ^{6}Li configuration, against $r = \beta^*_x/\beta^*_{x,HA}$ with the vertical plane held at high acceptance: the tagged fraction ε for pots following the 10σ envelope (dashed); the luminosity $L/L_{HA} = 1/\sqrt{r}$ (grey); their product, the tagged yield relative to the high-acceptance luminosity (thick), with the optimum marked and the yield, best-bin 5σ floor and shape-term significance per year annotated; and the yield if both planes were de-squeezed by r (thin). The envelope is a rectangle $10\sigma_h \times 10\sigma_v$ in angle with the dispersive term added horizontally, which is why the optimum sits far above the circular 1/e point of Table 2; B = 50 GeV^{-2}. With the pots fixed at the aperture measured in the ePIC geometry ε peaks at 1.2×10^{-5} over r and is stated in each panel rather than drawn. The dash-dotted line is the IR-8 secondary focus at our interpolated ≈ 20%, drawn as if the second IR delivered the high-acceptance luminosity; HA and HD mark the Yellow Report optics.

The reference tag of the projections. Figures 1 and 5 and §6.2 are drawn for a near-beam envelope of 0.20 GeV — ε_{tag} = 13.5% (9–20% for B ∈ 40–60 GeV^{-2}), the documented limit of coherent ^{3}He tagging at IP6 [22] — at the high-acceptance luminosity, which no IP6 optics provides: conservative in the tagged fraction against the 25–37% of the tagging optics and the ≈ 20% of IR-8, but 2.5×, 6.9× and 3.9× above what the tagging optics yields at the three configurations. They stand as the physics case for a tag, not as an IP6 projection.

The configuration we propose. A lithium tagging optics at 18 × 137.5 GeV/u with the horizontal β* raised ≈ 89× over the high-acceptance setting, the vertical plane untouched, and the Roman Pots inserted horizontally to ≈ 0.12 mrad (3.52 mm at the pot plane if the measured lever of 29.97 m survived the de-squeeze; the de-squeezed transfer matrix is one of the inputs requested below) — or, at the lowest energy, 5 × 40.8 GeV/u with β^*_x ≈ 46× (≈ 42 m for a high-acceptance β^*_x of 0.9 m) and pots at 0.36 mrad, 6.99 mm; the middle configuration asks 0.19 mrad and 4.08 mm. Against the 53.6, 31.1 and 26.7 mm the pots insert to today — a 2.4 cm base plus the per-energy offsets of the geometry file — that is an approach a factor 7.6–7.7 closer at every configuration, and it is the half of the lever the β* de-squeeze cannot supply by itself — run as a separate running configuration, as the Yellow Report foresees its high-acceptance optics, for a share of the year with which the yields above scale (Table 4, row 8) — the size of that share is a programme decision rather than a measurement, and is priced in plans/07 WP2 — while the inclusive measurement takes the high-divergence optics. It is a change to the interaction-region lattice, on the electron side as well, and to the pot insertion; whether a horizontal de-squeeze of that size is compatible with the crab cavities and keeps the far-forward transport parallel-to-point is part of what is asked. From the machine it needs the lithium divergence and the high-acceptance β* per configuration and the far-forward transfer matrix at the pot planes for the de-squeezed lattice; from the detector, the pot approach and the charge identification below; and its cost — a factor 6.8–12.8 in luminosity for the duration of the store — is the number to weigh against a second interaction region. Whichever provides the tag, one degeneracy remains: a Roman-Pot hit measures A/Z, and an intact ^{6}Li, an α and a d at the same velocity share trajectory and timing. The second-detector design answers it with a Z^2-scaling Cherenkov counter behind the pots [19]; for IP6 no charge-discrimination study exists, though the AC-LGAD readout digitizes an 8-bit charge over four planes and the α + d pair separates by hit topology [21].

6.2 · Rate model

$$\sigma_{\rm coh} = f_{\rm coh}(x)\,\sigma_{\rm DIS}, \qquad f_{\rm coh}(x) = \frac{f_0}{1+(x/x_{\rm coh})^2}, \qquad f_0 = 0.04\ [0.02\text{–}0.08]$$

No diffractive prediction exists for a light nucleus (the lightest published case is Ca), so the f_0 band brackets HERA ep diffraction (10–15% of DIS [23]) and heavy-nucleus saturation estimates (20–25% [24]); parameterizing f_{coh} in Bjorken x rather than x_P is optimistic for high-mass diffraction, which the band is meant to cover. The scale $x_{coh} = 0.01$ marks where the coherence length $l_c = 0.105$ fm/x_P falls below the ^{6}Li size, and the slope B = 50 GeV^{-2} (band 40–60) follows from $B = \langle r^2\rangle/3$ with the matter radius (the charge radius gives ≈ 57 [25]; gluonic distributions may be wider [26]), valid below the form-factor minimum at $|t| \approx 0.31$ GeV^2 [27,28]. One year produces 1.2×10^8 coherent events, of which the 0.20 GeV tag keeps 1.7×10^7 (1.7×10^8 in ten years); the best super-bin, $x \in [0.0013, 0.0025]$, $Q^2 \in [1, 1.7]$ GeV^2, holds 1.75×10^6 and gives $\delta\hat{A} = 1.8\times10^{-3}$ (one year) and 0.6×10^{-3} (ten years) on the amplitude per unit P_{zz}, i.e. 5σ floors at 0.9% and 0.3%; at $P_{zz} = 0.6$ the visible modulations are 0.6× these.

6.3 · The modulation and its two mechanisms

The cos 2φ′ modulation of the tagged yield carries two mechanisms. The *deformation* term is scaled from the polarized-deuteron colour-glass calculation of Mäntysaari et al. [10], the only published calculation of polarization-dependent azimuthal coefficients in coherent diffraction on any polarized nucleus (all forward citations checked; the group's subsequent work is unpolarized [29,30]). Their coefficients show a_2 growing linearly in |t| with $a_2(m = 0) = -2a_2(m = \pm1)$, from the 6% D-wave modulating the diffractive slope. For a fill with populations p_m the ensemble coefficient is proportional to P_{zz}:

$$a_2(t;P_{zz}) = -\frac{P_{zz}}{4}\,\varepsilon_{B0}\,B\,|t|, \qquad \langle a_2\rangle_{\rm tag} = a_2\big(p_{T,\rm cut}^2 + 1/B\big) = 0.036 \;\; [0.018\text{–}0.059]$$

Two conventions matter. The anchor's Eq. (9) expands $d^2\sigma/d\Phi d|t| \propto 1 + 2\Sigma_n a_n e^{in\Phi}$ with Φ the azimuth of the recoil relative to the polarization axis in the γ*d frame, so the cos 2Φ coefficient is $2a_2$: the $\langle a_2\rangle$ quoted here — 0.036 at $P_{zz} = 0.6$, i.e. 0.060 per unit P_{zz}, the footing of the 5σ floors — and injected in Figure 5(d) understates the deformation modulation by a factor two, and the coherent numbers of this paper are conservative by ×2. And the modulation is one of the *recoil* azimuth $\varphi_t - \varphi_S$, not of the electron azimuth used as the axis of Figure 5(d); the reconstructed-level analysis fits both azimuths [16].

$\varepsilon_{B0} \equiv \Delta B_0/B$ is scaled from the deuteron value of 0.21 by two indicators that disagree: the relative quadrupole deformation $Q/\langle r^2\rangle$ of ^{6}Li (0.012) is about one fifth of the deuteron's (0.063) and opposite in sign, while the α–d asymptotic D/S ratio is disputed — a transfer measurement gives η = +0.0003(9), consistent with zero, older sub-Coulomb analyses favour η ≈ −0.01, and the ab initio no-core shell model with continuum gives −0.027, the deuteron's own 0.0256 in magnitude [31]. We adopt $\varepsilon_{B0} \in -(0.04\text{–}0.13)$ with default −0.08, leaving near-zero deformation open; the D/S indicator alone would allow up to the deuteron's −0.21, and the 5σ floors cover detection or exclusion anywhere in that range. The band's width is genuine theoretical uncertainty: even exact few-body calculations miss the ^{6}Li quadrupole cancellation by a factor ≈ 3 [32], and the gluonic deformation need not share the cancellation of the charge sector. The sign flip relative to the deuteron is a testable prediction ($Q(^6\mathrm{Li}) < 0$), and tensor modulations of comparable size are established at these |t|: the elastic e-d $T_{20} = -(0.1\text{–}0.3)$ [33] and HERMES $A_{zz}(d) \approx -1\%$ at x ≈ 0.01 [34], the latter attributed to coherent double scattering [35–37]. The flat gluon-transversity term is separately bounded at 3×10^{-3}–10^{-2} (at $P_{zz} = 1$) from the bare lattice φ transversity [6] divided by the ~10× NPLQCD deuteron suppression [7], with the maximal Drell–Yan scenario [5] as the ceiling. $\langle a_2\rangle$ averages the m-state populations with equal weights; weighting them by their m-dependent rates lowers it by up to 27%, a one-sided shift inside the ε_{B0} band.

The two mechanisms separate by shape: deformation is linear in t, sign-flipped relative to the deuteron and expected to weaken toward small x_P (JIMWLK evolution washes it out [30]), while exotic glue is flat in t and persists at small x, where it connects to the elliptic-gluon programme [38]. In a two-component fit $a_2(t) = c_{def}\cdot|t| + a_g$ per x_P bin, the ten-year floor of 0.3% per unit P_{zz} lies at or below the flat term's band and an order of magnitude below the deformation term, so both components are expected to be resolved for the 0.20 GeV tag. Incoherent breakup is m-state-blind in this mechanism [10]: it dilutes the modulation but cannot mimic it. And deformation-driven modulations, the polarized-deuteron imaging signal, are bounded on ^{6}Li by its quadrupole moment, Q = −0.0806(6) fm^2 ≈ $Q(^7\mathrm{Li})/50$ [39]: a sizable flat-in-t cos 2φ on ^{6}Li cannot be a shape effect, and the mechanisms entangled for the deuteron separate in the ^{6}Li/d comparison — the null test unavailable to the deuteron.

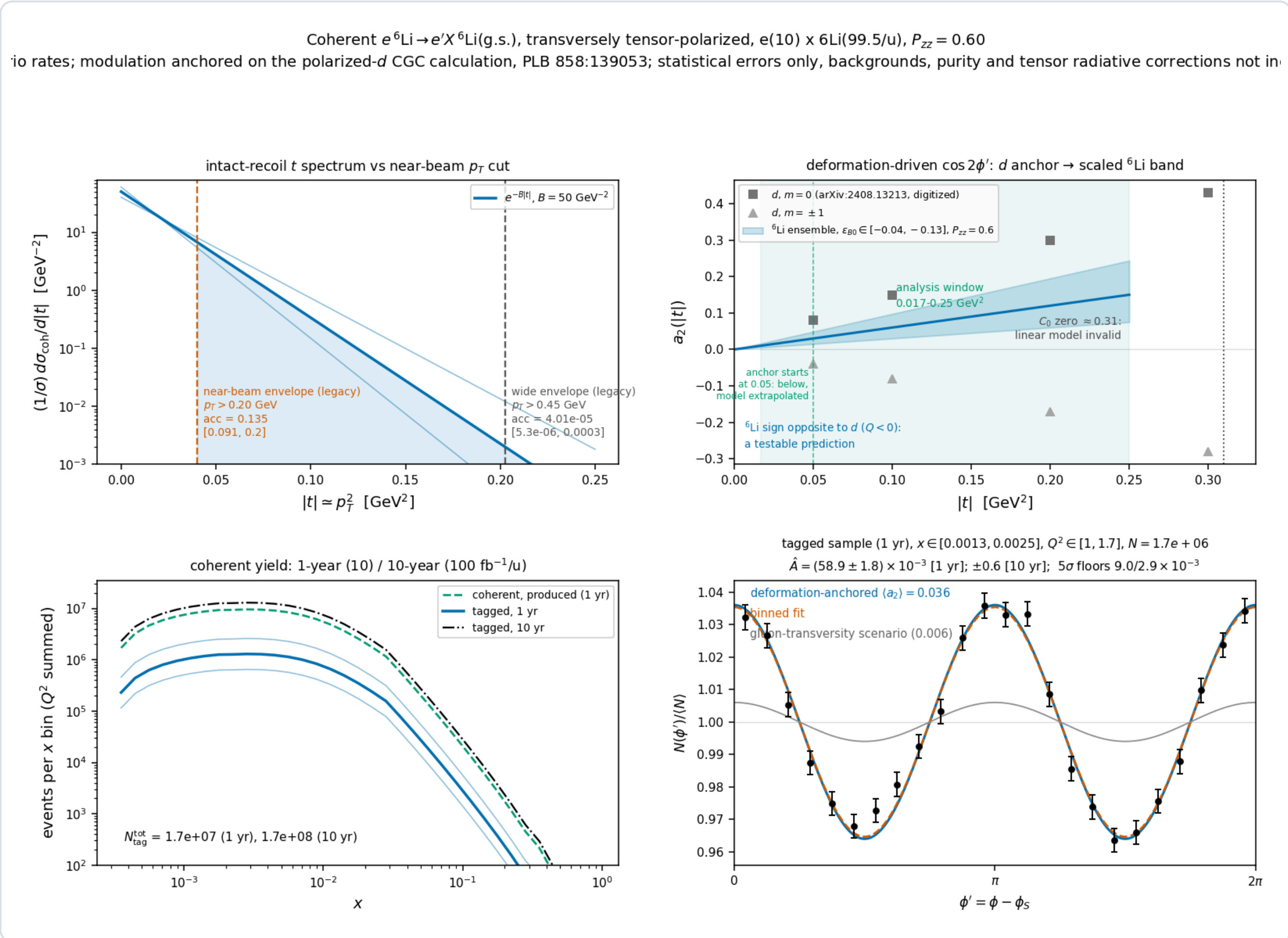

Figure 5 — the coherent channel e ^{6}Li → e′ X ^{6}Li(g.s.) at e(10 GeV) × ^{6}Li(99.5 GeV/u), P_{zz} = 0.60, for a 0.20 GeV near-beam envelope. (a) Recoil t spectrum with the slope band and the 0.20 and 0.45 GeV cuts (13.5% and 4×10⁻⁵); the panel's optics labels are the proton optics the two cuts were first derived from, neither of which is an available ^{6}Li optics (§6.1). (b) Digitized deuteron a_2 per m state [10] against the scaled ^{6}Li ensemble band (sign flip predicted), with the analysis window 0.017–0.25 GeV² — the reconstructed binning of money plot 6R (Report 2 §5.2), whose lower edge is a choice of the analysis above the tagging-optics aperture floor of 0.0079–0.0132 GeV² and not the floor itself — the marker at |t| = 0.05 GeV² where the digitized anchor begins, below which the band, and the a_t truth of the four reconstructed bins whose t_{ref} falls there, are the linear model extrapolated, and the form-factor-zero exclusion. (c) Tagged yield versus x for both programmes with the f_0 band. (d) φ′ pseudo-data (one-year statistics) at ⟨a_2⟩ = 0.036 with the flat gluon-transversity scenario (grey); extraction Â ± 1.8×10⁻³ (one year), ± 0.6×10⁻³ (ten years) per unit P_{zz}.

That the analysis closes once a tag exists has been demonstrated with the reconstructed-level chain [16]: the recoil is transported to the pots with divergence smearing and an aperture, the spin-state ratio is formed in the two azimuths and fitted with an acceptance-weighted template basis in which the pot geometry's own ⟨cos 2β⟩ cancels, and reflection-forbidden sin terms serve as null columns. At the tagging optics of §6.1 at 5 × 40.8 GeV/u — the Yellow Report divergence with the horizontal β* de-squeezed, pots following the 0.36 × 3.8 mrad envelope, tagged fraction 36%, 2.3×10⁶ tagged events per year at 1/6.8 of the high-acceptance luminosity, 5.5×10⁵ in the lowest of seven |t| bins over 0.017–0.25 GeV² — the fit recovers the deformation coefficient a_t = 0.0418 ± 0.0063, 0.0593 ± 0.0052, 0.0714 ± 0.0052, 0.0973 ± 0.0037, 0.1209 ± 0.0050 and 0.1535 ± 0.0089 against 0.046, 0.060, 0.073, 0.092, 0.121 and 0.150 injected — unbiased in the five bins that hold more than 10⁵ recoils, low by 2.5% over two hundred draws in the sixth, which holds 5.3×10⁴ (Report 2 §7), and, in the seventh, with 1.3×10⁴ events, 0.1166 ± 0.0186 against 0.183: the low-count bias of the bin-wise ratio, which the ten-year sample removes at 0.1799 ± 0.0060 and which is now removed outright by the acceptance-profiled likelihood that carries the same templates, returning 0.1796 ± 0.0192 on that same draw and a mean of 0.1814 over twenty one-year pseudo-experiments against 0.183 injected, where the ratio's is 0.1211 — resolves the pot roll to 15 mrad through the null columns in the 0.05–0.08 GeV² bin and 44 mrad in the lowest, and recovers the flat term in that lowest bin to ± 0.0022: a single pseudo-experiment returns 0.0099, and the twenty-experiment ensemble a mean of 0.0105 with a spread of 0.0018 against 0.010 injected, so the estimator is unbiased at these counts (Report 2 Table 5). Taken together the seven bins measure the flat term to $\delta a_e = 0.00121$ per year at 5 × 40.8, and to 0.00111 and 0.00074 at the other two configurations, against 0.00207 for the four-bin 0.05–0.25 GeV² window this analysis used until 2026-08-28 (Report 2 §7).

6.4 · Backgrounds

BACKGROUND	WHY IT FAKES THE TAG	HANDLE
α + d breakup (threshold 1.474 MeV; the 2.186 MeV 3^+ state, Γ = 24 keV, decays ~100% to α+d [40])	Both fragments at beam rigidity, on the trajectory and velocity of an intact ^{6}Li: tracking and time-of-flight blind; only dE/dx (∝ Z^2: 9/4/1) or the two-hit topology separates	Roman-Pot charge discrimination (open item #19, §6.1). Fallback: two-component |t| fit (coherent $e^{-50\|t\|}$ against a ~10× flatter incoherent slope; crossover at 0.05–0.2 GeV², inside the window [26,41], with the three bins added below 0.05 GeV² in 2026-08-28 lying below the crossover, so the fit now has a coherent-dominated lever arm it did not have); e+Pb benchmark 80–99% incoherent rejection [42]
Nucleon-emission breakup: α+p+n (3.70 MeV), ^{3}He+t (15.79 MeV) [40]	A fragment could pass unnoticed	p → off-momentum detectors (R = 0.50), n → ZDC, ^{3}He → Roman-Pot window (R = 0.75): vetoable; inefficiency is second order
^{6}Li* 3.5629 MeV (0^+, T = 1, Γ = 8.2 eV) [40]	Particle-stable (α+d parity-forbidden); γ-decays to a real A = 6, Z = 3 track	Isoscalar exchange cannot drive T = 0 → 1; the boosted 0.1–0.4 GeV γ lands mostly in the B0 calorimeter (photons ≥ 50 MeV, 5.5–20 mrad [20]), tail in the ZDC; the 5.366 MeV 2^+ T = 1 state always emits a nucleon. Every T = 1 state is γ- or nucleon-vetoable; only T = 0 states feed the blind α+d channel
Bethe–Heitler / QED Compton	Intact nucleus by construction	Calculable (t dependence = charge form factor squared [27,28]); subtract and reuse as an acceptance candle; its charge-quadrupole cos 2φ is < 10^{-3} for |t| < 0.15 GeV² [28,32]
Beam-gas / halo; e′-side fakes	Accidental beam-rigidity tracks; π / pair-symmetric e′ at high y	Non-colliding bunches, ~30 ps timing, vertex association; standard DIS e′ identification plus (e′, recoil) kinematic consistency

Table 3 — backgrounds to the coherent tag, ranked. The α + d channel is the one that decides the purity.

7 · Systematics and assumptions

Two extraction systematics survive the spin-state ratio: the polarimetry scale $\delta P_{zz}/P_{zz} \approx 3\%$, common with A_{zz} and with tensor polarimetry still requiring dedicated development (Lamb-shift or Breit–Rabi-type polarimeters) [8], and radiative corrections for tensor observables, which have not been calculated. The unpolarized part is now bounded rather than assumed: collinear initial-state radiation cannot fake a cos 2φ′ — the covariant azimuth is invariant under $k \to (1-z)k$ for a massless target — it is common to the two fills and cancels in the ratio, and it leaves $x = Q^2_e/(s\, y_\Sigma)$ exact, moving only the Q^2_e label of the bin by $1/(1-z)$; the migration that label carries biases $\mathcal{A}$ by +0.5 to +1.2% at the four sweet spots (≤2.9% once the generator window is opened below Q^2 = 0.15 GeV² and the low-Q^2 feed-in saturates, ≤0.25% behind a HERA-style $E - p_z$ window the chain holds in reserve rather than applies), against the ≤5% gate this programme set for its simulation letter (Report 2 §7). The tensor-sector correction is a different object and no unpolarized study stands in for it; the unpolarized QED tail is calculable and doubles as calibration. The detector nuisances have been measured rather than argued, holding the response fixed and varying one thing with common random numbers: a ±1% electron energy scale moves $\mathcal{A}$ by 0.3–0.9% at the four bins, a hadronic-resolution mismatch (0.30 generated, 0.25 corrected) by 0.5–1.4%, a 1% residual hadronic-scale error by 0.1–0.6%, all above the 0.06–0.28% Monte Carlo floor and below the bin-centering model dependence of §5.3, while an η tilt of the electron-identification efficiency (≤ 0.03%) and the Yellow Report's η-dependent calorimeter table (≤ 0.1%) are unresolved at that floor. In interpretation, a ΔΔ-isobar admixture can mimic Δ at leading twist [12] and conventional double scattering generates the low-x b_1 [35–37]; the Q^2 lever arm and the ^{6}Li/d comparison discriminate against both. For the coherent channel, an error in the unpolarized cos 2(α − β) harmonic u_2 reaches the flat term at first order through $a_t\, \delta u_2 \langle \cos 2\beta \rangle$ — a ZEUS-1σ δu_2 [43] moves a_e by 3–9% at the tagging optics of §6.1 (it was 20% under the slot geometry assumed before the aperture was measured) — and a fill-to-fill change of the pot cutout enters through the acceptance's own ⟨cos 2β⟩: at the tagging cutout a 10^{-3} change of the horizontal half-width moves a_t by 0.5–9.1% and a 10^{-2} one by 3.5–83.3% over the seven |t| bins at 5 × 40.8 (0.2–31.2% and 2.4–298.6% over the three configurations), the sensitivity climbing steeply towards the lowest bin, where the horizontal edge cuts a steep part of the spectrum, and where the earlier slot geometry amplified 10^{-3} to 19% in every bin [16]. The pots' envelope must therefore be stable to 10^{-3} between the spin states — which holds the four bins above 0.05 GeV² to 1.9% or better, the three below it needing 10^{-4} for the same — or the states must alternate bunch by bunch. Table 4 lists the assumptions and what a wrong one costs.

#	ASSUMPTION	IMPACT IF WRONG
1	A near-beam tag for the intact recoil exists — a tagging optics with pots that follow, or IR-8 (§6.1); the projections take a 0.20 GeV envelope at the high-acceptance luminosity, which no IP6 optics provides; the tagging-optics optimum (a 0.09–0.11 GeV horizontal envelope at 1/6.8–1/12.8 of the luminosity) yields 2–7× fewer events (§6.1)	No tag, no coherent channel; the inclusive measurement is unaffected
2	Coherent fraction f_0 = 0.04 [×2/÷2], in Bjorken x (§6.2); no diffractive model for a light nucleus exists	Yields scale linearly; δA as $1/\sqrt{f_0}$
3	Bag moment ported to per-nucleon ^{6}Li with dilution 1/3; x shape ours (§4)	Overall Δ scale; the literature brackets both directions [7,8]
4	ε_{B0} scaled from the deuteron by two indicators disagreeing by ×3 (§6.3)	Deformation amplitude anywhere in the band, which every plot carries
5	No event-by-event Z identification in the pots; purity from the |t|-shape fit (§6.4)	Z identification would improve purity; an incoherent slope model that is off grows the fit systematic
6	Linear $a_2(t)$ for |t| < 0.2 GeV² only; the form-factor zero at 0.31 GeV² is excluded, not modelled; and the digitized deuteron anchor begins at |t| = 0.05 GeV², so in the four reconstructed bins whose t_{ref} falls below it the linear shape is extrapolated past its lowest anchor point	The upper tagged window carries unmodelled structure, and the four lowest |t| bins — including the lowest bin of the window published until 2026-08-28 — rest on an extrapolation of the anchor rather than on an anchor point
7	Toy structure functions with grid backends and a placeholder R; $g_2 = g_2^{WW}$ (§3.2); the target-mass kernel on by default since 2026-08-29 (§2)	Rates to ×1.5; shapes are scenarios; the R shift of §3.2 applies to every Δ; the target-mass term is a $(1 + \gamma^2)$ factor on $A_\parallel$ alone and reaches no Δ either way, so the flip moves nothing in this report
8	All luminosity in transverse-tensor ^{6}Li fills at P_{zz} = 0.6, the m = 0-rich fill at −1.2 with equal purity, alternating bunch by bunch (§3.3), and all of it at this measurement's own optics — spin state, far-forward optics (§6.1) and isotope are three layers of one run-plan choice, and every projection here takes the whole year in its own (§3.1)	N scales with the run-plan share and the amplitude with P_{zz}; the coherent and inclusive reaches are alternatives, not a sum; without alternation the bias of §3.3
9	Impulse approximation; no final-state interactions; tensor radiative corrections unknown	FSI distorts the recoil spectrum at larger |t|; the collinear-ISR migration is bounded at ≤2.9% of Δ (§7, Report 2 §7), the tensor-sector correction is not
10	A calorimeter noise floor of 50 MeV on the hadronic $E - p_z$ sum, calorimeters to |η| = 3.7 (70–85% of it captured at the four bins) and a hadronic scale calibrated from the simulation in reconstructed bins (§5.2) [16]	The purities of Table 1 degrade in proportion to the noise; at 100 MeV the y ≈ 0.01 bins are lost at the mid configuration. Uncalibrated, the forward escape is a 13–27% scale bias on y and the purities fall to 0.42–0.68; a 1% residual moves $\hat{A}$ by 0.1–0.6%; the ePIC nominal reach of 4.0 recovers a quarter of the escape without changing δy/y

Table 4 — the assumptions behind the projections, ranked by leverage.

8 · Summary and outlook

A transversely tensor-polarized ^{6}Li beam at the EIC measures $\Delta(x,Q^2)$ in inclusive DIS at 21–44σ per (x, Q²) bin in one year under the moment-constrained scenario, and at 3–11σ per bin for a flat amplitude of 3×10^{-3} per unit P_{zz} even at P_{zz} =

0.3, discriminating between interpretations of the bag moment by the x dependence and extracting $x\Delta(x,Q^2)$ to 2–9% in the best bins. The result survives reconstruction with the spin-state ratio and a hadronic-method y, at the price of bin purities that the calorimeter noise floor controls. The measurement makes three demands on the machine that are specific and answerable: tensor polarization delivered as alternating m = ±1-rich and m = 0-rich bunches, transverse at the interaction point; a cos 2φ′ acceptance and a pot envelope stable to 10^{-4} between the spin states, or the bunch-by-bunch alternation that makes any relative-luminosity error a constant; and a tensor polarimetry scale at the few-per-cent level.

The coherent channel carries the same physics with a null test the deuteron cannot offer. It is also a requirement rather than a projection: the intact ^{6}Li cannot be tagged at IP6 with any published optics, at any energy, and what recovers it is the tagging optics of Figure 4 or the secondary focus of the second interaction region. The tagging optics we propose (§6.1) — 18 × 137.5 GeV/u with the horizontal β* raised ≈ 89× and pots at 0.12 mrad, or 5 × 40.8 GeV/u at ≈ 46× and 0.36 mrad, the vertical plane untouched, as a separate running configuration — recovers a 33–37% tag at 1/9.5–1/6.8 of the luminosity: a 9–11σ measurement per year of the shape term with its predicted sign flip against the deuteron, and 5σ on a 1% exotic-glue term in three years — 2.6 at the top configuration, 3.0 at the low one; a second interaction region's secondary focus would deliver 4–6× more at the same luminosity. The highest-leverage inputs from outside the programme are, on the machine side, a light-ion optics with β* chosen for tagging and a charge-identification study of the pot readout; on the theory side, a rerun of the IP-Glasma polarized-deuteron setup with an α–d cluster density, which would replace both ε_{B0} indicators and the f_0 band at once, a lattice Δ moment for a light nucleus, and radiative corrections for tensor observables.

References

[1] R. L. Jaffe, A. Manohar, *Nuclear gluonometry*, Phys. Lett. B 223 (1989) 218.

[2] P. Hoodbhoy, R. L. Jaffe, A. Manohar, *Novel effects in deep inelastic scattering from spin-one hadrons*, Nucl. Phys. B 312 (1989) 571.

[3] J. Maxwell et al., *Search for exotic gluonic states in the nucleus*, JLab LOI12-16-006, arXiv:1803.11206.

[4] E. Sather, C. Schmidt, *Size and scaling of the double-helicity-flip hadronic structure function*, Phys. Rev. D 42 (1990) 1424.

[5] S. Kumano, Q.-T. Song, Phys. Rev. D 101 (2020) 054011, arXiv:1910.12523; Phys. Rev. D 101 (2020) 094013, arXiv:2003.06623.

[6] W. Detmold, P. E. Shanahan, Phys. Rev. D 94 (2016) 014507 [Erratum: Phys. Rev. D 95 (2017) 079902], arXiv:1606.04505; W. Detmold, D. Pefkou, P. E. Shanahan, Phys. Rev. D 95 (2017) 114515, arXiv:1703.08220.

[7] F. Winter et al. (NPLQCD), Phys. Rev. D 96 (2017) 094512, arXiv:1709.00395.

[8] G. Atoian et al. (EPIOS), *Realizing the scientific program with polarized ion beams at EIC*, Phys. Rev. C 113 (2026) 060501, arXiv:2510.10794.

[9] C. Peng and J. Zhou, *Deep inelastic scattering on polarized lithium: the physics case for ^{6}Li and ^{7}Li beams at the Electron-Ion Collider*, Report 0 of this series (2026).

[10] H. Mäntysaari, F. Salazar, B. Schenke, C. Shen, W. Zhao, *Spatial imaging of polarized deuterons at the Electron-Ion Collider*, Phys. Lett. B 858 (2024) 139053, arXiv:2408.13213.

[11] W. Cosyn, B. Roldan Tomei, A. Sosa, A. Zec, Eur. Phys. J. A 61 (2025) 83, arXiv:2410.12764.

[12] M. Nzar, P. Hoodbhoy, *Estimation of the double helicity flip deuteron structure function*, Phys. Rev. D 45 (1992) 2264.

[13] R. Abdul Khalek et al., *EIC Yellow Report*, Nucl. Phys. A 1026 (2022) 122447, arXiv:2103.05419 (Tables 10.1–10.2, §10.1); C. Peng and J. Zhou, *The EIC and ePIC for a polarized-lithium programme: beam and detector parameters, their provenance, and the lithium case derived*, Report 3 of this series (2026).

[14] K. Abe et al. (E143), Phys. Lett. B 452 (1999) 194, hep-ex/9808028.

[15] A. Bacchetta, M. Diehl, K. Goeke, A. Metz, P. J. Mulders, M. Schlegel, JHEP 02 (2007) 093, hep-ph/0611265.

[16] C. Peng and J. Zhou, *Simulation and reconstruction chain of the inclusive and coherent cos 2φ measurements with a tensor-polarized ^{6}Li beam at the EIC: assumptions, methodology and closure*, Report 2 of this series (2026).

[17] S. Maple (ePIC), *Tracking and inclusive DIS reconstruction with the ePIC detector at the EIC*, seminar, Birmingham, 11 December 2024; ATHENA Collaboration, JINST 17 (2022) P10019, arXiv:2210.09048, §3.1; M. Arratia, D. Britzger, O. Long, B. Nachman, Nucl. Instrum. Meth. A 1025 (2022) 166164.

[18] A. Jentsch, Z. Tu, C. Weiss, Phys. Rev. C 104 (2021) 065205, arXiv:2108.08314.

[19] Second EIC detector / IR-8 far-forward design: *Precision studies of QCD in the low energy domain of the EIC*, arXiv:2211.15746 (§VIII); BNL LDRD 23-050 closeout, arXiv:2602.04636.

[20] M. Pitt, PoS DIS2024 (2025) 259, arXiv:2409.02811; E.-C. Aschenauer et al., Phys. Rev. D 112 (2025) 036010, arXiv:2503.05908.

[21] C. Peng and J. Zhou, *Near-beam tagging of the far-forward lithium recoils at ePIC: what the aperture is worth, what the optics is worth, and whether superconducting nanowires help*, Report 4 of this series (2026).

[22] W. Chang, E.-C. Aschenauer, A. Jentsch, A. Kumar, Z. Tu, Z. Yin, *Opportunities for imaging light nuclei with a second interaction region at the EIC*, Phys. Rev. D 113 (2026) 032018, arXiv:2511.05638.

[23] H1 Collaboration, Eur. Phys. J. C 72 (2012) 2074, arXiv:1203.4495.

[24] T. Lappi, Nucl. Phys. A 830 (2009) 403c, arXiv:0907.4588; H. Kowalski, T. Lappi, C. Marquet, R. Venugopalan, Phys. Rev. C 78 (2008) 045201, arXiv:0805.4071.

[25] I. Sick, arXiv:1505.06924.

[26] STAR Collaboration, Phys. Rev. Lett. 128 (2022) 122303, arXiv:2109.07625.

[27] L. R. Suelzle, M. R. Yearian, H. Crannell, Phys. Rev. 162 (1967) 992.

[28] G. C. Li, I. Sick, R. R. Whitney, M. R. Yearian, Nucl. Phys. A 162 (1971) 583.

[29] H. Mäntysaari, H. Roch, B. Schenke, C. Shen, W. Zhao, Phys. Rev. D 114 (2026) 014068, arXiv:2605.00454.

[30] H. Mäntysaari, P. Singh, Eur. Phys. J. C 85 (2025) 1449, arXiv:2411.14934.

[31] K. D. Veal, C. R. Brune, W. H. Geist et al., Phys. Rev. C 60 (1999) 064003 (transfer measurement); G. Hupin, S. Quaglioni, P. Navrátil, Phys. Rev. Lett. 114 (2015) 212502 (NCSMC).

[32] R. B. Wiringa, R. Schiavilla, Phys. Rev. Lett. 81 (1998) 4317, nucl-th/9807037.

[33] M. Bouwhuis et al. (NIKHEF), Phys. Rev. Lett. 82 (1999) 3755; D. Abbott et al. (JLab t20), Phys. Rev. Lett. 84 (2000) 5053; D. M. Nikolenko et al. (VEPP-3), Phys. Rev. Lett. 90 (2003) 072501.

[34] A. Airapetian et al. (HERMES), Phys. Rev. Lett. 95 (2005) 242001, hep-ex/0506018.

[35] N. N. Nikolaev, W. Schäfer, Phys. Lett. B 398 (1997) 245.

[36] K. Bora, R. L. Jaffe, Phys. Rev. D 57 (1998) 6906, hep-ph/9711323.

[37] J. Edelmann, G. Piller, W. Weise, Phys. Rev. C 57 (1998) 3392.

[38] Y. Hatta, B.-W. Xiao, F. Yuan, Phys. Rev. Lett. 116 (2016) 202301.

[39] N. J. Stone, *Table of nuclear electric quadrupole moments*, IAEA INDC(NDS)-0833 (2021) and At. Data Nucl. Data Tables 111–112 (2016) 1.

[40] D. R. Tilley et al., *Energy levels of light nuclei A = 5, 6, 7*, Nucl. Phys. A 708 (2002) 3 (TUNL).

[41] LEPS Collaboration, coherent φ photoproduction on ^{4}He, arXiv:1711.01095.

[42] W. Chang et al., Phys. Rev. D 104 (2021) 114030, arXiv:2108.01694.

[43] ZEUS Collaboration, Nucl. Phys. B 816 (2009) 1, arXiv:0812.2003 (leading-proton azimuthal asymmetries A_{LT}, A_{TT}).

Appendix A · Revision history

DATE	REVISION
2026-08-29	Verification pass over the round. The horizontal-to-vertical lever ratio is quoted per configuration, 6.3 at 10 × 100 and 10.2 at 18 × 275, in place of a single factor 9 that is neither. The Z/A = 0.5 lattice edge is quoted as the 1.61 mrad the code applies (48 mm / 29.81 m), with the measured first hit at 1.60 on +x and 1.70 on −x. The closure bound of §5.2 is stated as 2.3 standard errors, the value Report 2 also carries. The de-squeeze factors and the pot approach angles are stated once, where they are derived; the cross-reference in Figure 5's caption points at Report 2 §5.2, where money plot 6R is tabulated, and the last revision row at §6.1 rather than §5.2. The dateline names Report 4. §6.1 and Figure 4's caption carry the fixed-pot ceiling the 2026-08-28 aperture re-measurement leaves: with the pots held at the measured silicon the tagged fraction peaks at 1.2×10^{-5} rather than staying below 10^{-6}, the horizontal edge at 18 × 137.5 having moved from 1.03 to 0.53 mrad; the conclusion that the far-forward side must move with the optics is unchanged, and so are the yields, which are quoted for pots that follow the envelope. The ceiling is stated as a maximum rather than as a bound, in the text and in Figure 4's own panel annotation, which now reads max $\varepsilon = 1.2 \times 10^{-5}$ over r in place of a $< 1 \times 10^{-5}$ the printed 1.23×10^{-5} does not satisfy (`tagging_optics.py`). Table 1's spot-4 one-year bar is 2.9×10^{-4}, §5.2's low-configuration resolution 0.39 and its 7R stand-in error at $Q^2 = 1.14$ GeV² 2.4×10^{-3}, and the calibrated amplitude closes within 7% of the true-bin value rather than 6%. This appendix now runs newest first as Reports 2–4 do. Later the same day the per-configuration far-forward transport became the baseline in place of the single 18 × 275 lever, and every number in this report that stands on it moved: the de-squeeze factors are 46.5 / 164.1 / 89.3 and the luminosity 1/6.8 / 1/12.8 / 1/9.5, the tagged yields 2.4 / 2.4 / 6.2×10^6 a year, the tagged fractions 0.37 / 0.25 / 0.33, three to six years to 5σ on a 1% exotic-glue term, and a second interaction region worth 4–10× rather than 3–8×. The 2026-08-28 row above was restated on those levers rather than annotated, so it reads as the current numbers and not as the numbers of that day. §6.1 now states the pot insertion the optics asks for in millimetres — 6.99, 4.08 and 3.52 mm against the 53.6, 31.1 and 26.7 mm inserted today — so the ask is a requirement list and not an open lever. §2's target-mass clause follows the generator's default, which was flipped on the same day: `InclusiveKernel` carries the exact finite-γ $A_{\parallel}$ unless asked not to, and the $(1 + \gamma^2)$ factor it computes reaches no Δ either way.
2026-08-28	Table 2's pot aperture re-measured in the current ePIC geometry (`epic-main` at git 9aaa2969; plans/09 B1, Report 4 §6): $\lvert\theta_x\rvert \gtrsim 2.50$ / 1.51 / 0.53 mrad against the September-2024 2.0 / 1.35 / 1.03, so ε_{tag} at the aperture is 9×10^{-10} / 2×10^{-19} / 1×10^{-5}, and the transport that images an IP angle onto the pot plane is measured per configuration for the first time, R_{12} = 19.24 / 21.25 / 29.97 m against R_{34} = — / 3.35 / 2.93 m. The conclusion of §6.1 stands — the largest entry is still 10^{-5} — but which constraint delivers it changes hands: the silicon binds at 5 × 40.8 and the beam at 18 × 137.5, the reverse of the earlier reading at both ends. The 5 × 41 row is conditional on which of that ring setting's two lattices a ^{6}Li fill runs in, worth ×0.64 in the aperture. The ε_{tag}-envelope column and every projection of §6 are unchanged, the tagging optics depending on the pot dispersion only through D/R_{12}, whose re-measured 18 × 275 value lands within 0.6% of the one it was priced with.

2026-08-28	§3.1 completed with the two halves of the run-plan law it had stated only for statistical errors: a luminosity quoted as a reach in fb^{-1}/u is invariant under the share, the years needed to accumulate it go as one over it, and what is not statistics follows neither — the full one-year Δ bars of Report 2 [16] carry the spread over unfolding priors, which no luminosity shrinks. Assumption row 8 of Table 4 and the configuration proposed in §6.1 name the three layers the share divides — spin state, far-forward optics and isotope — and plans/07 WP2 prices the options. No projection moves.
2026-08-28	The coherent channel of §6.3 and §7 restated on the seven $\|t\|$ bins over 0.017–0.25 GeV^2 adopted in Report 2 §7, which replace the four of 0.05–0.25 GeV^2 every earlier revision quoted. At 5 × 40.8 the window now holds 82% of the tagged sample instead of 28% and measures the flat exotic-glue term to $\delta a_e = 0.00121$ per year instead of 0.00207; every coherent number of §6.3 is re-measured on it, as are the two coherent systematics of §7 — a ZEUS-1σ δu_2 moves a_e by 3–9% rather than 7–9%, and a 10^{-3} cutout change moves a_t by 0.5–9.1% rather than ≤ 0.8%, the added low bins being the least sensitive to the first and the most to the second, so the 10^{-3} envelope requirement is now edge-of-window dependent. Assumption row 6 of Table 4 gains the lower-side caveat that the digitized deuteron anchor begins at $\|t\| = 0.05$ GeV^2, and Figure 5(b) shades the analysis window and marks that start. No inclusive projection moves.
2026-08-28	The target-mass term of the longitudinal $A_\parallel$ implemented behind `InclusiveKernel(target_mass=True)`, default off, and bounded where §2 and assumption row 7 of Table 4 had asserted it small without a number: $\gamma^2 \le 0.0057$ at the four bins of §3.4, where the exact kernel collapses to a multiplicative $(1 + \gamma^2)$ on $A_\parallel$ worth ≤ 0.6% — ≤ 0.5% at the top configuration and ≤ 2.5% at the low one, whose $Q^2 = 1.14$ GeV^2 spot sits at $x = 0.089$ — and ≤ 0.097 anywhere the $W^2 \ge 10$ GeV^2 cut allows (`evgen/scripts/target_mass_bound.py`). No Δ reported here is built from $A_\parallel$, and no published number moves.
2026-08-28	The unpolarized radiative correction moved from an exclusion to a bound (plans/07 WP4; Report 2 §7). §7 and assumption row 9 now carry the measured collinear-ISR migration — +0.5 to +1.2% of Δ at the four sweet spots in the published generator window and ≤2.9% once that window is opened below $Q^2 = 0.15$ GeV^2, against the programme's ≤5% gate — in place of the earlier ≤1.1%, which quoted one response seed and the truncated window. The tensor-sector correction remains uncalculated and is stated as such. §6.3 gained the acceptance-profiled likelihood of Report 2 §4.3, which removes the low-count bias of the sparsest $\|t\|$ bin outright (mean 0.173 ± 0.005 against 0.180 injected on the same twenty draws, where the ratio gives 0.114); the ratio remains the default and no published error bar moves.
2026-08-28	Numbers of §5.2 and §7 restated after the review of the reconstruction toolchain (Report 2 Appendix A, `docs/code_review_2026-08-28.md`): the hadronic-scale calibration is now keyed on the reconstructed (x, Q^2) rather than the event's true cell, and the PYTHIA library was regenerated without PYTHIA's default $\hat{m} \ge 4$ GeV floor, so the calibrated purities of Table 1 read 0.56 / 0.59 / 0.64 / 0.75 (D = 1.00 / 1.06 / 0.95 / 0.98) instead of 0.52 / 0.59 / 0.59 / 0.76; the coherent systematics of §7 are quoted at the tagging optics of §6.1 (δu_2 → 7–9% on a_e, a 10^{-3} cutout change → ≤ 0.8% on a_t) instead of the superseded slot geometry. No projection of §5.1 or §6 moved.
2026-08-27	Rewritten as a paper on the γ-matched beam energies (^6Li at 40.8 / 99.5 / 137.5 GeV/u, plans/10). All numbers restated from the current scripts: sweet spots at $x = 0.011$–0.14, amplitudes $(0.44$–$0.95)\times10^{-2}$ against $\delta A = (1.4$–$4.5)\times10^{-4}$, the reconstructed-level Table 1, the R-model shifts (+18, +19, +8, −4%), the bin-centering scan, 1.2×10^8 coherent events and 1.7×10^7 tagged for the 0.20 GeV envelope. The coherent channel restated against the Yellow Report divergences and the measured pot aperture (Table 2): it does not survive at IP6 with any published optics, and the figures are presented for a tagging optics or IR-8 rather than as an IP6 projection; the pre-2026-08-27 6R and measured-aperture results, computed at the superseded rigidity-scaled energies (2.7×10^6 tagged at 50 GeV/u; the low-energy survival at 20.5 GeV/u), are withdrawn and replaced by the analysis closure at a 0.73 mrad approach. Assumptions condensed to Table 4; references numbered in citation order. Later the same day: the tagging optics priced (Figure 4, `tagging_optics.py`) — luminosity $\propto 1/\beta^*$ per plane against the tagged fraction, a horizontal-only de-squeeze with the vertical plane untouched, dispersion at the pots and the parallel-to-point transport assumption stated, the pots following or fixed — and the IP6 lithium tagging configuration proposed in §6.1 with its cost.
2026-08-26	Development run 8: the PYTHIA 8 hadronic final state; the Roman-Pot aperture measured in the ePIC geometry; the published R1998 fit behind an r_func hook; the forward-folded bin-centering alternative.
2026-08-25	The spin-state-sorted ratio as the estimator of the measurement; the near-beam cut stated as angular; the b_1, b_2 sign convention; the inversion corrected; the $O(\gamma^2)$ b_1 leakage; far-forward parameters updated.
2026-08-24	The anchored coefficient's convention ($2a_2$); the reconstruction-chain report created as the reconstructed-level companion.
2026-08-17	Dedicated fact checks of the Sather–Schmidt moment usage and of the far-forward detection chain; phase-space figure with the analysis binning; display mathematics typeset at build time.
2026-08-11	Second audit round (formulas against the literature, implementation, references, extraction logic); money plot 7, the extracted Δ as structure-function data points, added.

`2026-08-10`	First version (development run 5): money plots 5 and 6 with the moment-constrained $\Delta(x,Q^2)$ and separate one-year / ten-year projections, and the physics note with a verified bibliography; first audit round the same day.

Generated from the PolarizedLithiumSim repository: `evgen/scripts/money_cos2phi.py`, `money_cos2phi_coherent.py`, `money_delta_extraction.py`, `phase_space_bins.py`; reconstructed level `money_cos2phi_reco.py` (with `--y-source hfs`, `--unfold-scan`, `--syst-scan`), `hfs_resolution.py`, `coherent_optics_scan.py`, `tagging_optics.py`, `nearbeam_aperture_scan.py`, `nearbeam_reach_gain.py`; models in `polli_fastsim/delta_models.py` and `evgen/polligen/coherent.py`; assembled by `reports/build_report.py`. Every command is in docs/reproduction_manual.md. All rates use scenario inputs — bands, not predictions; external numbers are verified against primary sources. Target venue: INT programme on polarized ion beams at the EIC, March 22 – April 2, 2027.